Weather Dataset Assignment

20 ProblemStatementsusingNumPy&Pandas|Real-LifeWeatherDataset

Dataset	Real-Life Weather Data — 365 days, 5 Indian Cities
Columns	Date, City, Temperature_C, Humidity_%, Wind_Speed_kmh, Rainfall_mm, Pressure_hPa, Visibility_km
Libraries	NumPy, Pandas
Total Problems	20

Problem 1: What is the average temperature across all cities?

Code:

```
avg_temp = np.nanmean(df['Temperature_C'].values)

print(f'Average Temperature = {avg_temp:.2f} C')
```

Output: Average Temperature = 27.56 C

Problem 2: Find the maximum and minimum wind speed recorded.

Code:

```
max_wind = np.max(df['Wind_Speed_kmh'].values)

min_wind = np.min(df['Wind_Speed_kmh'].values)
```

Output: Max Wind Speed = 79.7 km/h | Min Wind Speed = 0.0 km/h

Problem 3: How many missing values exist in each column?

Code:

```
missing = df.isnull().sum()

print(missing)
```

Output: Temperature_C: 15 | Rainfall_mm: 10 | All other columns: 0

Problem 4: What is the total rainfall recorded for each city?

Code:

```
city_rainfall = df.groupby('City')['Rainfall_mm'].sum().round(2)

print(city_rainfall)
```

Output: Bangalore:1729.2 | Chennai:2425.9 | Delhi:2405.1 | Kolkata:2833.0 | Mumbai:1662.4

Problem 5: What is the standard deviation of humidity?

Code:

```
std_humidity = np.std(df['Humidity_%'].values)

print(f'Std Dev of Humidity = {std_humidity:.2f} %')
```

Output: Standard Deviation of Humidity = 22.50 %

Problem 6: Filter all days where temperature exceeded 40 degrees C.

Code:

```
hot_days = df[df['Temperature_C'] > 40]

result = hot_days[['Date', 'City', 'Temperature_C']]
```

Output: 50 days found with Temperature > 40 C

Problem 7: What is the median pressure across all records?

Code:

```
median_pressure = np.median(df['Pressure_hPa'].values)

print(f'Median Pressure = {median_pressure:.2f} hPa')
```

Output: Median Pressure = 1002.00 hPa

Problem 8: Which city has the highest average humidity?

Code:

```
city_humidity = df.groupby('City')['Humidity_%'].mean()

top_city = city_humidity.idxmax()
```

Output: Chennai — Avg Humidity = 67.43%

Problem 9: How many rainy days (Rainfall > 0) were there?

Code:

```
rainy_days = df[df['Rainfall_mm'] > 0].shape[0]

print(f'Rainy Days = {rainy_days}')
```

Output: 146 rainy days

Problem 10: Normalize temperature values between 0 and 1 using NumPy.

Code:

```
temp_vals = df['Temperature_C'].dropna().values

normalized = (temp_vals - temp_vals.min()) /

(temp_vals.max() - temp_vals.min())
```

Output: First 5 normalized values: [0.193, 0.411, 0.701, 0.135, 0.129]

Problem 11: What is the correlation between Temperature and Humidity?

Code:

```
corr = df[['Temperature_C', 'Humidity_%']].corr()

print(corr)
```

Output: Correlation between Temperature and Humidity = -0.06 (weak negative)

Problem 12: Fill missing Temperature values with the column mean.

Code:

```
mean_temp = df['Temperature_C'].mean()

df['Temperature_C'] = df['Temperature_C'].fillna(mean_temp)
```

Output: Missing values in Temperature_C after fill = 0

Problem 13: What is the month-wise average rainfall?

Code:

```
df['Month'] = df['Date'].dt.month

 monthly_rain = df.groupby('Month')['Rainfall_mm'].mean().round(2)
```

Output: Jan:14.53 | Feb:36.62 | Mar:38.03 | Apr:40.03 | May:38.04 | ... | Dec:26.99

Problem 14: Find days with visibility below 5 km (foggy days).

Code:

```
foggy = df[df['Visibility_km'] < 5]

result = foggy[['Date', 'City', 'Visibility_km']]
```

Output: 75 foggy days found

Problem 15: Compute the variance of wind speed using NumPy.

Code:

```
wind_var = np.var(df['Wind_Speed_kmh'].values)

print(f'Variance of Wind Speed = {wind_var:.2f}')
```

Output: Variance of Wind Speed = 544.48

Problem 16: Sort the dataset by Rainfall in descending order and show top 5 records.

Code:

```
top_rain = df.sort_values('Rainfall_mm', ascending=False).head(5)

print(top_rain[['Date', 'City', 'Rainfall_mm']])
```

Output: Highest: Kolkata — 148.0 mm on 2023-09-24

Problem 17: What percentage of days had no rainfall?

Code:

```
no_rain_pct = (df['Rainfall_mm'] == 0).sum() / len(df) * 100

print(f'{no_rain_pct:.2f}% days had no rainfall')
```

Output: 57.26% days had no rainfall

Problem 18: Add a new column 'Heat_Index' = Temperature * Humidity / 100.

Code:

```
df['Heat_Index'] = np.round(
df['Temperature_C'].values * df['Humidity_%'].values / 100, 2)
```

Output: Heat_Index column added successfully

Problem 19: Which city recorded the highest single-day rainfall?

Code:

```
max_rain_idx = df['Rainfall_mm'].idxmax()
row = df.loc[max_rain_idx, ['Date', 'City', 'Rainfall_mm']]
```

Output: Kolkata — 148.0 mm on 2023-09-24

Problem 20: Compute the 90th percentile of temperature using NumPy.

Code:

```
p90 = np.percentile(df['Temperature_C'].values, 90)
print(f'90th Percentile of Temperature = {p90:.2f} C')
```

Output: 90th Percentile of Temperature = 41.42 C

All 20 Problem Statements Completed Successfully!